# Hybrid Model for Detecting Alcoholism from EEG Signals using Deep Learning and Machine Learning

Gherson Torres[1] and Wilfredo Ticona[1,2]

[1] Universidad Tecnológica del Perú, Lima, Perú
U17104172, C19404}@utp.edu.pe
[2] Universidad ESAN, Lima, Perú
wmamani@esan.edu.pe

**Abstract.** Alcoholism is a public health problem with significant neurological consequences, as it alters the brain's electrical activity and impairs cognitive function. The objective of this research was to implement hybrid models based on machine learning and deep learning to detect alcoholism from electroencephalogram (EEG) signals. A structured four-stage methodology was applied: Dataset acquisition; Preprocessing of EEG images (resizing, normalization, and labeling); Deep Learning (feature extraction using the following architectures: ViT, PiT, CNN, MobileNetV2, BotNet, and MLPMixer); Machine Learning (classification using the following algorithms: SVM, RF, and k NN); and evaluation. The best results were obtained with the ViT+SVM hybrid model, with an accuracy of 98.38%, precision of 96.91%, recall of 100%, and an F1-Score of 98.43%. In conclusion, the results show that the hybrid model based on ViT+SVM can detect alcoholism from changes in the EEG. This study is of great importance in preventing irreversible damage and identifying individuals with alcoholism so that they can receive timely clinical treatment.

**Keywords:** Alcoholism, EEG, Deep Learning, Machine Learning, Hybrid model, Vision Transformer, Hyperparameters tuning.

## 1 Introduction

Alcoholism is currently one of the most significant public health problems worldwide. It is not only about the number of deaths but also about the social, family, and economic impact it generates [1]. The World Health Organization (WHO) estimates that nearly 3 million people die each year from causes related to alcohol consumption, which corresponds to approximately 5.3% of all global deaths. Excessive alcohol consumption is linked to more than 200 diseases and injuries, such as liver cirrhosis, pancreatitis, various types of cancer, cardiovascular diseases, and psychiatric disorders [1] and [2]. In Latin America, the situation is more worrying. The Pan American Health Organization indicates that alcohol is directly responsible for some 85,000 deaths annually and contributes to more than 300,000 additional deaths, in a context where consumption is often higher than the global average [3]. On the other hand, in the case of Peru, the Ministry of Health has reported that between 1.5 and 2 million adults have problematic alcohol

consumption and that around 85% of the population has consumed alcohol at some point in their lives. This means that a large part of the population is at risk of developing dependence, confirming the magnitude of the problem in the country [4].

The diagnosis of alcohol use disorder is traditionally based on clinical interviews and standardized questionnaires [5]. These methods have several limitations. On the one hand, they depend heavily on the patient's honesty, and many people deny their actual consumption out of fear, shame, or ignorance. On the other hand, the interpretation of symptoms and medical history can vary among specialists, leading to differences in diagnoses. This process can be time-consuming, costly, and difficult to implement on a large scale [4] and [5]. Given these limitations, more methods are needed. One option is to use measurements of brain electrical activity, known as electroencephalograms (EEGs), which record how the brain functions electrically. Frequent alcohol consumption directly affects brain function: it alters communication between nerve cells and changes the rhythm of brain waves. These alterations make it possible to recognize patterns linked to the problem and transform EEG readings into concrete data that help confirm the diagnosis of alcoholism [5]. However, the signals obtained from the EEG are complex and often full of noise, making them difficult to interpret directly. To solve this problem, signal processing and machine learning techniques can be applied to transform these signals into images using the continuous wavelet transform [6], [7], create scalograms in the time and frequency domains. This representation enables the use of deep learning architectures, such as CNNs (e.g., VGG, ResNet, and EfficientNet) and vision transformers (ViT, PiT, and DeiT). While CNNs focus on detecting local patterns, Vision Transformers establish global relationships between frequencies and moments through the self-attention mechanism [8].

This capability is particularly relevant for detecting distributed patterns associated with alcoholism. However, its application in EEG is still limited and has been little evaluated in clinical contexts. For this reason, this study suggests a specific approach: using wavelet-transformed EEG scalograms as the basis for various deep learning models, focusing on Vision Transformer (ViT) and Pooling Vision Transformer (PiT), and analyzing their performance in comparison with CNN and other architectures, both in their “pure” modality and in combination with machine learning architectures such as SVM, Random Forest, and k-NN [6]. The main objective is to investigate the feasibility of developing a robust, accurate, and efficient model that facilitates the automatic identification of alcoholism from EEG, providing an objective tool that supports diagnosis and complements clinical criteria. This proposal is aligned with the epidemiological reality at the global, Latin American, and Peruvian levels and supports the development of automatic detection systems based on advanced machine learning and deep learning techniques [2], [6] and [8].

## 2 Related Work

In their study, Zhang et al. [5], pointed out that one of the main problems in detecting alcoholism early is that many people do not honestly acknowledge their consumption, which limits the effectiveness of traditional assessments. Therefore, they propose a more objective method based on brain signals obtained through EEG. In their research, they transformed EEG signals into two-dimensional heat maps of the brain and used

transfer learning techniques to differentiate between individuals at risk of developing alcoholism and healthy individuals. To this end, they tested 21 combinations of traditional feature extractors together with normal classifiers, as well as 84 combinations using pre-trained deep convolutional neural networks (such as DenseNet, VGG, MobileNet, and Xception) as feature extractors, together with classifiers such as SVM, k-NN, Random Forest, and MLP. The findings revealed that the most effective combination was MobileNet together with an SVM with an RBF kernel, achieving 95.33% precision, 95.68% accuracy, 95.24% F1-Score, and 95.00% recall, surpassing conventional methods that only use EEG features by up to 8 percentage points. In summary, the study demonstrates that transforming EEG signals into heat maps and using pre-trained deep models allows for the creation of a diagnostic support tool with an accuracy close to 95%, presenting a more objective and effective option for the early detection of predisposition to alcoholism.

In the work of Anuragi and Singh Sisodia [7], the authors propose a hybrid approach to detecting alcohol consumption from EEG that combines an information extraction stage with a supervised classification phase. First, they transform the signals using the flexible analytical wavelet transform, which allows them to describe brain activity in the time-frequency domain and capture subtle patterns associated with alcoholism that are not visible with simpler methods. Based on these descriptors, they construct feature vectors that are then fed into different machine learning classifiers, specifically SVM, Random Forest, and k-NN, evaluating various configurations until the most effective one is found. Their results show that the model based on SVM with RBF kernel offers the best performance, achieving nearly 98.36% accuracy, 98% recall and precision, and an F1-score of approximately 98%. These metrics exceed those obtained by other classifiers and approaches that use less elaborate features by several percentage points, reinforcing the idea that a good combination of rich feature extraction (such as wavelets) and robust ML classifiers can offer very high and stable performance in the detection of alcoholism using EEG.

In this study, Wu et al [9] ,indicated that traditional methods based on interviews and clinical scales do not allow for a completely objective assessment of drug and alcohol addiction. Therefore, they propose the use of EEG signals to directly capture the changes in brain activity produced by these substances. Their goal is to develop a deep learning model called DSCnet capable of differentiating between healthy individuals and those with addiction based on a hybrid representation of EEG and feature learning from multiple perspectives, such as local, global, dynamic, and static information. The results show that DSCnet achieved an accuracy of 85.11% in the drug addiction set, with 85.13% precision, 85.12% recall, and 85.12% F1-score. Similarly, the UCI set achieved an accuracy of 84.56%, precision of 84.73%, recall of 84.56%, and F1 score of 84.63%, outperforming other classical approaches and classifiers. Overall, these results indicate that the model is correct in more than eight out of ten cases and maintains a good balance between correctly detecting patients with addiction and avoiding misclassifying healthy subjects, making it a promising tool for diagnostic support and clinical follow-up.

Finally, Mumtaz et al. [10] point out that questionnaires and interviews to detect alcohol use disorder (AUD) can be unreliable, so they propose using resting EEG as an objective measure of brain connectivity. Their approach is based on a measure of functional connectivity called synchronization likelihood (SL), calculated from 19

electrodes and reduced by feature selection to retain only relevant information. They used EEG from 30 patients with AUD and 30 healthy subjects, training SVM, Naive Bayes, and logistic regression models. The results showed that SVM performed best, with 98% accuracy, 99.9% sensitivity, and 95% specificity, whereas the other models achieved accuracy between 91.7% and 93.6%. In conclusion, the study demonstrates that SL-based functional connectivity features can serve as objective markers of AUD and enable the development of a highly accurate automatic screening tool that is useful as an aid to clinical diagnosis.

## 3 Methodology

The proposed methodology is structured in five main phases: (1) data acquisition, (2) preprocessing, (3) application of deep learning models, (4) application of the hybrid approach (ML, DL), (5) model evaluation. The methodology is visualized in Fig. 1.

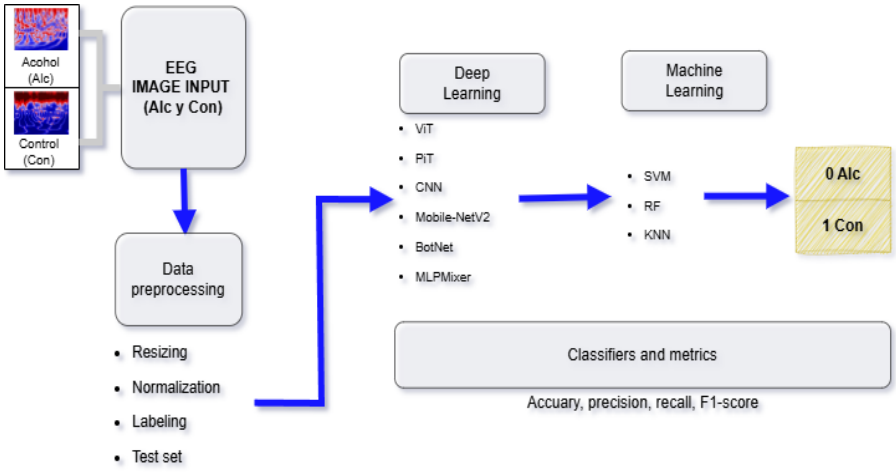


**Fig. 1.** Proposed methodology.

### 3.1 Data acquisition

For this research, we used the dataset called "wavelet-transformed EEG data for alcohol detection," available on Kaggle: https://www.kaggle.com/datasets/muhammedjunayed/wavelet-transformed-eeg-data-for-alcohol-detection/data. This dataset consists of signals captured by 61 brain channels (EEG), which were visually converted into 924 scalograms using continuous wavelet transformation. These images are 200×172 pixels and allow us to see both the time and frequency of the signal simultaneously. They were divided into 468 alcoholism samples and 456 control samples.

### 3.2 Preprocessing

Preprocessing is divided into four stages. These stages prepare EEG scalogram images for use in deep learning models, ensuring consistent and adequate data for training.

- **Resizing:** Resizing adjusts the size of EEG scalogram images from their original resolution of 200×172 pixels to 224×224 pixels. This action is crucial since most deep learning models (ViT, PiT, and MobileNetV2) require inputs with fixed dimensions. Standardizing the size reduces memory consumption, speeds up processing, and makes it easier to work with pretrained models, allowing transfer learning without starting from scratch. This size adjustment ensures that all models receive images with uniform measurements, allowing them to identify real patterns [11].
- **Normalization:** Normalization helps keep input data within a controlled range, promoting numerical stability during the model training process, specifically in EEG scalogram images before they are fed into deep learning systems [11]. In this work, RGB images have pixel values between 0 and 255, and each is divided by $I_{\text{norm}} = I/255$, so applied to bring them to the continuous range $[0, 1]$ before training models such as ViT, PiT, CNN, and MobileNetV2 [11] and [12]. In this way, the models work with a stable and comparable numerical scale across the 924 images and can focus on the actual EEG patterns rather than artificial differences in intensity.
- **Labeling:** Labeling is the step where each EEG scalogram image receives a numerical "label" indicating whether it belongs to an alcoholic or a control subject. Images from control subjects (Con) are assigned the label 0, whereas those from alcoholic subjects (Alc) receive the label 1.
- **Test set:** The test set determines whether deep learning models generalize correctly rather than memorizing the training data. Of the 924 EEG scalogram images, 80% (739 images) are used for training and 20% (185 images) for testing; the latter, which are not used during training, are reserved exclusively for calculating performance metrics (accuracy, precision, sensitivity, and F1-Score).

### 3.3 Application for Deep Learning Models

Six deep learning models were trained and evaluated on the set of EEG scalograms:

- **ViT (Vision Transformer):** This is a model that classifies images using a structure like that of a Transformer applied to sections of the image. The image is segmented into parts of constant size, each of which is converted into a linear representation, position representations are incorporated, and the resulting series of vectors is fed into a conventional transformer encoder. To perform the classification, the usual technique of including an additional "classification token" that can be learned in the sequence is used [12].
- **PiT (Pooling-based Vision Transformer):** This is a type of model derived from ViT that adds pooling phases between the transformer blocks. These phases gradually reduce the number of sections, allowing the system to first capture nearby details and then understand more complex overall connections. Thus, PiT combines the total attention of transformers with a layered organization like convolutional networks [13].

- **CNN Convolutional Neural Network):** Este This is a model that uses time and frequency representations, such as electroencephalograms, to convert brain signals into a visual form. These images allow the CNN to analyze the EEG in a similar way to how it would analyze a traditional image. Convolutional layers identify local patterns related to energy distribution across various frequency bands, while pooling layers reduce dimensionality and highlight the most important areas. Finally, fully connected layers combine the information obtained to classify brain states, such as identifying alcohol consumption [14].
- **MobileNetV2 (Mobile Network Version 2):** This is a type of convolutional neural network designed to be lightweight and efficient. It uses units called inverted residuals that facilitate the extraction of key image features with a reduced number of parameters. When applied to time-frequency EEG images, such as scalograms, MobileNetV2 recognizes local patterns related to different frequency bands, providing a favorable balance between accuracy and low computational requirements. Therefore, it is an ideal model for evaluating its effectiveness compared with more elaborate architectures [15].
- **BotNet (Bottleneck Transformer):** This model combines convolutional neural networks (CNN) and Transformer-type self-attention mechanisms. In its first layers, the model acts like a conventional CNN, extracting local patterns from the image. In the final layers, spatial convolution is replaced by multi-head self-attention, which allows information from different areas of the image to be linked. This combination is suitable for analyzing time and frequency images, such as EEG scalograms, where relevant patterns may be distributed throughout the visual representation [16] .
- **MLPMixer (Multilayer Perceptron Mixer):** This is a model that analyzes images differently from traditional networks. Instead of using attention mechanisms, it first segments the image into small parts known as patches and then processes this information solely through multilayer perceptrons (MLPs). Its structure alternates between two simple processes: one that facilitates the exchange of information between the different patches of the image and another that integrates the information within each patch. Thanks to this approach, the model can directly discover spatial relationships and meaningful patterns. In the case of time-frequency images, such as EEG scalograms, the MLP-Mixer is used as a reference model to determine whether simpler structures can capture valuable information for classification [17].

### 3.4 Application of the hybrid approach

In this phase, the six deep learning models were used in combination with three machine learning classifiers (SVM, RF, KNN).

- **SVM (Support Vector Machine):** This is a classifier that seeks to separate two groups by finding a boundary that maximizes the distance between them. After deep learning models transform each image into a feature vector, the SVM finds the best boundary that separates the alcoholic subjects from the controls, maintaining the largest possible margin between the two groups. Thanks to this approach, SVM is usually a stable and accurate method [18].
- **RF (Random Forest):** This is a classifier that makes the final decision by merging the results of multiple decision trees. Once the deep learning models transform each

EEG image into a feature vector, each tree autonomously examines these data and provides a prediction. The final class is established through a majority voting process, which provides the model with greater stability and makes it less sensitive to noise, particularly when working with high-dimensional features produced by deep networks [19].

- **KNN (K-Nearest Neighbors):** This is a very intuitive machine learning model. Its main idea is simple: data are classified according to which data they most closely resemble. In deep learning models, they are first used to transform EEG scalograms into feature vectors that summarize the most relevant information. Subsequently, k-NN compares these vectors and classifies each subject as alcoholic or control according to the predominant class among the most similar neighbors. In this way, k-NN acts as a simple and interpretable classifier, used mainly as a reference to evaluate the quality of the learned representations [20].

### 3.5 Model evaluation

After training, each model (deep and hybrid) was evaluated on an independent set of EEG scalograms from tests. This allowed us to verify their actual ability to distinguish between alcoholic and control subjects. Four key metrics were used to measure performance: accuracy, precision, sensitivity (recall), and F1-Score, to obtain a comprehensive view of the effectiveness and predictive power of each model.

## 4 Results

This section presents the most relevant results of the proposed methodology. The parameters of each model were adjusted to optimize its performance.

### 4.1 Hyperparameters of machine learning classification algorithms

- **Support Vector Machine (SVM):**
  C: 0.1, 1, 10 | kernel: linear, Rbf | gamma: 0.01, scale.
- **Random Forest (RF):**
  n_estimators: 100, 200, 300 | max_depth: none, 20 | min_samples_leaf: 1, 2.
- **K-Nearest Neighbors (KNN):**
  n_neighbors: 5, 15, 25 | weights: uniform, distance | metric: cosine, euclidean.

### 4.2 Results of hybrid models

This section is based on a hybrid method that integrates deep learning with machine learning. The most notable results were achieved by merging two deep learning models that obtained the best ViT performances and associating them with machine learning (SVM).

**Table 1.** Results of hybrid models

| Feature Extractor | Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|---|
| **ViT** | **SVM** | **0.9838** | **0.9691** | **1.000** | **0.9843** |
| | RF | 0.9297 | 0.9263 | 0.9362 | 0.9312 |
| | KNN | 0.8757 | 0.8515 | 0.9149 | 0.8821 |
| PiT | SVM | 0.6054 | 0.6105 | 0.6170 | 0.6138 |
| | RF | 0.6108 | 0.6196 | 0.6064 | 0.6129 |
| | KNN | 0.6000 | 0.5909 | 0.6915 | 0.6370 |
| CNN | SVM | 0.9459 | 0.9286 | 0.9681 | 0.9479 |
| | RF | 0.9405 | 0.9109 | 0.9787 | 0.9436 |
| | KNN | 0.9459 | 0.9468 | 0.9468 | 0.9468 |
| MobileNetV2 | SVM | 0.6270 | 0.6404 | 0.6064 | 0.6230 |
| | RF | 0.5622 | 0.5684 | 0.5745 | 0.5714 |
| | KNN | 0.5568 | 0.5625 | 0.5745 | 0.5684 |
| BotNet | SVM | 0.5946 | 0.6234 | 0.5106 | 0.5614 |
| | RF | 0.6270 | 0.6344 | 0.6277 | 0.6310 |
| | KNN | 0.6432 | 0.6628 | 0.6064 | 0.6333 |
| MLPMixer | SVM | 0.6486 | 0.6593 | 0.6383 | 0.6486 |
| | RF | 0.6757 | 0.6771 | 0.6915 | 0.6486 |
| | KNN | 0.6541 | 0.6630 | 0.6489 | 0.6559 |

La Table 1 shows the performance of different hybrid models using various feature extractors. A notable result is the performance of the ViT extractor fed into an SVM; this configuration, implemented with a linear SVM ($C = 0.1$, Kernel = linear, Gamma = scale).This configuration achieves the best overall results, reaching 98.38% accuracy and perfect 100% recall, outperforming the other combinations. This suggests that ViT' ability to capture global relationships allows for the construction of a highly effective representation to distinguish between alcoholic subjects and controls, as confirmed by its F1-Score of 98.43%.

The confusion matrix is a tool that allows the effectiveness of classification models to be measured, as it facilitates the analysis of hits and errors, as well as promoting the constant refinement of the model [21].

Fig. 2 presents the results, showing the confusion matrix of the model with the highest metric values, showing both correct and incorrect predictions.

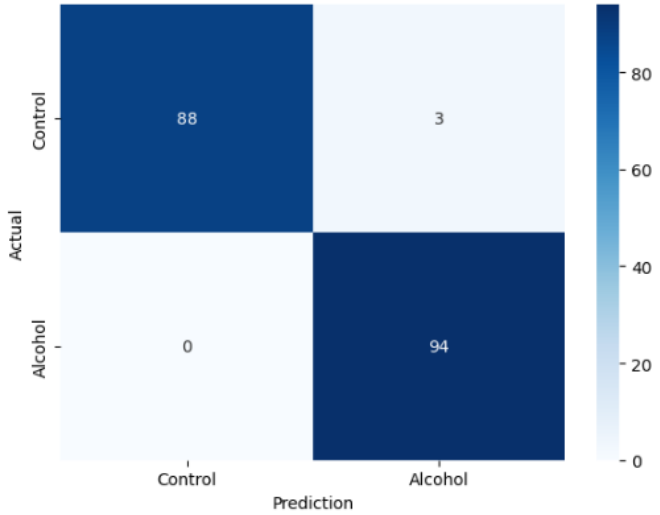


**Fig. 2.** Confusion matrix: ViT with SVM

The confusion matrix shown in Fig. 2 details the performance of the hybrid ViT + SVM classification model for distinguishing alcoholic subjects from controls using EEG images:

- **True Negatives (TN):** The model correctly classified 88 subjects in the control group as non-alcoholics, reflecting a good ability to recognize healthy individuals and avoid unnecessary diagnoses.
- **False Positives (FP):** Three control subjects were incorrectly labeled as alcoholics; in a clinical setting, these cases would represent additional alerts that would require verification by a specialist.
- **False Negatives (FN):** No false negatives were observed (0 cases), meaning that no alcoholic subjects were classified as controls, which is particularly relevant because the system does not omit cases of alcoholism within the analyzed set.
- **True Positives (TP):** A total of 94 alcoholic subjects were correctly identified as alcoholics, demonstrating the hybrid model's high capacity to detect the presence of alcoholism from EEG scalograms and incorrect predictions.

The ROC curve shows the relationship between sensitivity and false positive rate, analyzing how effectively the model distinguishes between sick and healthy people when changing thresholds [22].

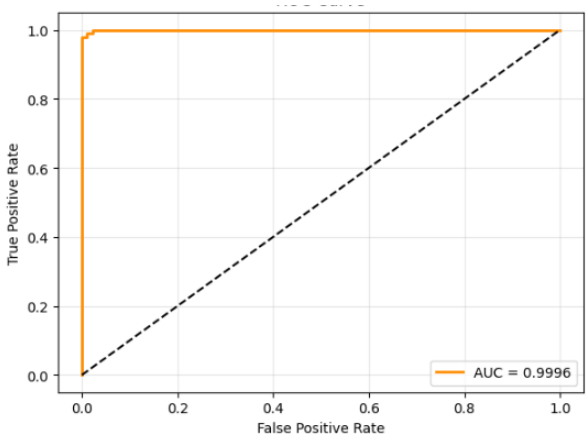


**Fig. 3.** ROC curve of the ViT+SVM model

Fig. 3 shows the orange curve rising almost vertically toward the upper left corner (AUC = 0.9996), indicating that the ViT+SVM model correctly identifies virtually all actual cases of alcoholism in EEG while maintaining minimal false positives, confirming its exceptional diagnostic performance.

## 5 Conclusions

This study addressed alcoholism as a public health problem with significant neurological consequences, given that it alters brain electrical activity and impairs cognitive function. It was emphasized that early detection of these changes in the EEG is key to preventing irreversible damage and supporting timely clinical interventions. Compared to traditional visual inspection methods, which are susceptible to fatigue and subjectivity, the use of artificial intelligence was proposed as a more efficient alternative for automatically differentiating between alcoholic subjects and control subjects.

Data acquisition was performed using Kaggle's “wavelet-transformed EEG data for alcohol detection” dataset, consisting of 924 scalograms obtained from EEG signals recorded on 61 channels. Each image, with a resolution of 200 × 172 pixels, simultaneously represents the temporal and frequency information of the signal and is labeled as Alcohol (468 samples) or Control (456 samples). During preprocessing, resizing, normalization, and labeling operations were applied to homogenize the images, reduce noise, and facilitate model learning while reducing the risk of overfitting.

The modeling stage focused on a hybrid model approach. First, different deep learning architectures were used as feature extractors (ViT, PiT, CNN, MobileNetV2, BotNet, and MLPMixer), responsible for automatically learning high-level descriptors from EEG scalograms. Subsequently, these feature vectors were used as input for machine learning classifiers (SVM, RF, and k-NN), exploring various combinations of hyperparameters for each algorithm. Among these configurations, the hybrid ViT + SVM methodology proved to be the most robust, achieving an accuracy of 98.38%, precision of 96.91%, sensitivity of 100%, and an F1-score of 98.43%, with no false negatives in the alcoholic class.

These findings show that the combination of deep extractors capable of capturing complex patterns in EEG signals with well-tuned classical classifiers allows for the construction of robust and clinically useful systems for the automatic identification of alcoholism. Although other combinations, such as those based on CNN with SVM or k-NN, offered competitive results, the overall analysis confirmed that the ViT + SVM hybrid model represents the most consistent solution among the architectures evaluated.

In conclusion, the results obtained support the feasibility of incorporating this type of hybrid approach into automated clinical systems capable of providing faster, more consistent, and reproducible diagnoses based on EEG, which is particularly useful in healthcare centers with limited resources.

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